Consensus-Guided Pseudo-Labeling via Heterogeneous Speech Ensembles for Unsupervised Speaker Verification

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***Abstract*—Unsupervised speaker verification aims to learn speaker-discriminative representations from unlabeled speech, but early clustering errors can be reinforced by iterative pseudo-label training. We propose consensus-guided pseudo-labeling (CGPL), which treats two heterogeneous speech pre-trained models as ensemble members and uses their aligned clustering agreement as an observable signal of assignment stability. Agreement is not assumed to guarantee label correctness. Instead, it is used to select an initial pseudo-labeled subset, after which highly similar clusters are merged to reduce over-segmentation. The selected samples are optimized with dynamic label-noise-aware supervision, whereas samples without cross-model consensus contribute through self-supervised contrastive learning with hard negative mining. On VoxCeleb2, the initial selection retains 69.19% of the utterances and increases pseudo-label F1 from 0.555 and 0.478 for the two input partitions to 0.692. After three training iterations, CGPL obtains EERs of 1.01%, 1.36%, and 2.53% on Vox1-O, Vox1-E, and Vox1-H, respectively. Speaker identities are excluded from pseudo-label construction and gradient-based optimization; they are used only for offline configuration analysis and result reporting. The findings therefore support cross-model consensus as a practical pseudo-label selection signal within the evaluated VoxCeleb protocol.**

***Index Terms*—Unsupervised speaker verification, ensemble learning, pseudo-label, label-noise-aware supervision, hard negative mining.**

I. Introduction

Speaker verification determines whether a test utterance belongs to a claimed speaker and is widely used in speech-based authentication. Deep neural network-based systems [1]–[5] have substantially improved speaker verification performance over traditional approaches [6], [7]. These advances, however, largely depend on large-scale speaker-labeled corpora. Although speech data can be collected at scale, reliable speaker identity annotations are often unavailable or require costly manual labeling. This gap between the availability of speech data and speaker identity labels has motivated the development of unsupervised speaker verification methods, which aim to learn speaker-discriminative representations without manual annotations.

Existing unsupervised speaker verification studies mainly follow two research directions. The first direction learns speaker representations through self-supervised objectives. Contrastive learning methods inspired by SimCLR [8] and MoCo [9] construct positive and negative speech pairs to learn speaker-discriminative representations without manual labels [10]–[14]. Self-distillation methods further improve representation stability by enforcing consistency across different augmented views while avoiding explicit negative-pair construction [22]–[30]. The second direction adopts iterative pseudo-label-based training, where speaker embeddings are clustered to generate pseudo-labels and the resulting speaker encoder is refined using these pseudo-labels [15]–[20]. By introducing speaker-level supervision from unlabeled data, these methods have narrowed the gap between unsupervised and supervised speaker verification systems. However, their performance remains highly dependent on the quality of pseudo-label supervision.

Despite recent progress, iterative unsupervised speaker verification is still challenged by unreliable pseudo-label generation and utilization. Clustering-based pseudo-labels inevitably contain incorrect assignments because some samples have ambiguous pseudo-label assignments under imperfect speaker representations. Moreover, utterances from the same speaker may be divided into multiple pseudo-label classes, resulting in speaker-cluster over-segmentation. When these imperfect pseudo-labels are directly used for model optimization, the introduced errors may be propagated and amplified during subsequent refinement iterations. Therefore, improving pseudo-label supervision and reducing the influence of residual label noise are important for stable iterative unsupervised speaker verification.

Previous studies have attempted to address pseudo-label-related challenges from different perspectives. Speech self-supervised pre-trained models, such as wav2vec 2.0, HuBERT, UniSpeech-SAT, and WavLM [31]–[35], provide transferable representations and have been shown to improve first-round pseudo-label generation [36]. Noise-resistant learning methods mitigate the influence of noisy supervision through regularized objectives, reliability-aware optimization, classifier adjustment, sample selection, or iterative label refinement [37]–[42]. Other approaches refine clustering results by filtering samples with low assignment confidence, pruning unreliable classes, or selecting samples using probabilistic criteria [36], [41], [42]. Although these methods improve representation learning, pseudo-label generation, or noisy-label optimization, they usually focus on individual stages of the learning process. In particular, reliable pseudo-label construction, residual noise mitigation, and effective utilization of samples with different reliability levels are rarely considered within a unified framework. As a result, samples without cross-model consensus are often discarded, while the remaining pseudo-labels may still contain residual errors.

To address these limitations, we propose a Consensus-Guided Pseudo-Labeling framework, termed CGPL, for iterative unsupervised speaker verification. CGPL formulates initial pseudo-label generation as heterogeneous ensemble inference: two speech pre-trained models act as complementary ensemble members, and their aligned clustering assignments provide an observable signal of cross-model assignment stability. The framework is designed to improve pseudo-label supervision while utilizing both consensus-selected and cross-view-disagreement samples in a unified noise-robust training pipeline. Specifically, CGPL uses cross-model clustering agreement to identify reliable pseudo-labels, introduces pseudo-label merging to alleviate speaker-cluster over-segmentation, and further applies noise-robust optimization strategies to handle residual pseudo-label errors and exploit cross-view-disagreement samples. The main contributions of this work are summarized as follows:

- We formulate initial pseudo-label generation from a heterogeneous ensemble-learning perspective and propose a consensus-guided pseudo-labeling framework for iterative unsupervised speaker verification. Cross-model assignment agreement is used as an explicit and operational stability criterion for pseudo-label selection.
- We develop a reliability-aware pseudo-label construction strategy that combines consensus-guided sample selection and pseudo-label merging to improve pseudo-label consistency while alleviating speaker-cluster over-segmentation.
- We propose a noise-robust training strategy for utilizing unlabeled samples with different reliability levels. Consensus-selected samples are optimized with dynamic label-noise-aware supervision, while cross-view-disagreement samples are further exploited through self-supervised contrastive learning with hard negative mining.
- Experiments on the VoxCeleb benchmarks show that the selected subset has higher initial pseudo-label quality than either input partition and that the complete framework achieves competitive verification performance after three training iterations under the reported configuration.

The remainder of this paper is organized as follows. Section II reviews related work on unsupervised speaker verification and noise-resistant learning. Section III formulates the problem of reliability-aware unsupervised speaker verification and provides an overview of the proposed CGPL framework. Section IV presents the proposed consensus-guided pseudo-label construction procedure, including cross-model agreement-based sample selection and pseudo-label merging. Section V describes the noise-robust utilization of consensus-selected and cross-view-disagreement samples, where selected pseudo-labeled samples are optimized with dynamic label-noise-aware supervision and the remaining samples are exploited through self-supervised contrastive learning. Section VI describes the experimental setup and evaluation protocol. Section VII reports the main comparisons, pseudo-label quality analysis, ablation studies, and iterative behavior. Section VIII discusses the interpretation and limitations of the evidence, and Section IX concludes the paper.

II. Related Work

A. Contrastive Learning for Speaker Representation

Contrastive learning has been widely adopted for unsupervised speaker representation learning because it can exploit unlabeled speech through pairwise similarity constraints. Inspired by frameworks such as SimCLR [8] and MoCo [9], previous studies [10]–[14] construct positive pairs from augmented views or segments of the same utterance and generally treat samples from different utterances as negative pairs. The resulting objectives encourage representations from the same utterance to remain close while separating those from different utterances.

The effectiveness of contrastive learning, however, depends on the validity of the negative-pair assumption. In large-scale unlabeled speech corpora, different utterances may originate from the same speaker and can therefore be incorrectly treated as negatives. Such false negatives introduce repulsive constraints between same-speaker samples and may degrade the learned representation [21]. Although existing studies improve contrastive speaker learning through data augmentation, margin-based objectives, positive-pair construction, and sample selection [10]–[14], contrastive learning alone does not provide explicit pseudo-label supervision and cannot directly address pseudo-label noise in iterative clustering-based systems.

B. Self-Distillation for Speaker Representation Learning

Self-distillation provides an alternative self-supervised paradigm that avoids explicit negative-pair construction. Methods based on teacher–student architectures enforce prediction consistency across multiple augmented views and have been applied to unsupervised speaker representation learning [22]–[26]. By removing manually defined negative pairs, these methods reduce the direct influence of false-negative sampling and improve representation stability.

Self-distillation has also been incorporated into iterative unsupervised speaker verification frameworks [27]–[30]. In these systems, the learned representations are clustered to generate pseudo-labels for subsequent model refinement. Although self-distillation improves the robustness of the underlying representations, it does not directly guarantee reliable clustering assignments. Ambiguous samples may still receive incorrect pseudo-labels, and utterances from the same speaker may be split into multiple pseudo-label classes. Therefore, additional mechanisms are needed to improve pseudo-label reliability during iterative training.

C. Pre-Trained Models for Unsupervised Speaker Verification

Speech self-supervised pre-trained models have become important feature extractors for unsupervised speaker verification. Models such as wav2vec 2.0 [32], HuBERT [33], UniSpeech-SAT [34], and WavLM [35] learn transferable representations from large-scale unlabeled speech and encode acoustic, phonetic, and speaker-related information that can support downstream speech tasks [44].

Previous studies [36], [49] have shown that features extracted from speech pre-trained models can improve clustering initialization and unsupervised speaker verification performance. In particular, these representations provide informative embeddings during the early stage of iterative pseudo-label training and can be combined with downstream speaker encoders to incorporate both pre-trained and task-specific information.

Nevertheless, improved representations do not necessarily lead to fully reliable pseudo-labels. Early clustering results may still contain samples with ambiguous pseudo-label assignments, and utterances from the same speaker may still be divided into multiple pseudo-label classes. Moreover, speech pre-trained models trained with different objectives may produce non-identical clustering partitions and error patterns. This diversity makes heterogeneous pre-trained models suitable ensemble members: consistent decisions provide a stronger reliability cue than the output of either individual model.

D. Ensemble Learning and Cross-Model Consensus

Ensemble learning combines predictions from multiple learners to obtain more reliable decisions than those produced by an individual learner [66], [67]. Its effectiveness depends not only on the competence of the members but also on diversity among their error patterns. In unsupervised learning, where reference labels are unavailable, agreement among heterogeneous members can serve as an observable indicator of decision stability, while disagreement indicates that the inferred assignment is model-dependent.

Most ensemble methods aggregate class posteriors or discrete predictions defined in a shared label space. Clustering ensembles are more challenging because cluster indices are permutation-invariant and independently generated partitions may contain different numbers of clusters. Therefore, cross-model agreement cannot be evaluated until the partitions have been aligned. CGPL addresses this issue by aligning two heterogeneous PTM-based partitions and using sample-level assignment agreement as an operational consensus criterion. The method does not assume that agreement proves label correctness; rather, it uses agreement to identify pseudo-labels that are stable across two independently learned representation spaces.

E. Iterative Frameworks for Unsupervised Speaker Verification

Iterative optimization has been widely used to reduce the performance gap between unsupervised and supervised speaker verification systems [15]–[20]. A typical framework alternates between two stages. First, speaker embeddings are extracted from unlabeled speech and clustered to generate pseudo-labels. Second, the pseudo-labels are used to train or refine a speaker encoder. The updated encoder then produces new embeddings for the next round of clustering and model optimization.

This iterative procedure enables speaker representations and pseudo-labels to improve jointly. Its effectiveness, however, is sensitive to the quality of the initial clustering results. Incorrect assignments generated in early iterations may be reinforced in later rounds, while speaker-cluster over-segmentation can provide inconsistent supervision for utterances from the same speaker. Consequently, improvements in representation learning alone may be insufficient when the pseudo-label supervision remains unreliable.

CGPL follows the iterative unsupervised speaker verification setting but focuses on pseudo-label reliability and sample utilization. Instead of treating pseudo-label construction and model optimization as independent stages, CGPL jointly considers reliable pseudo-label selection, speaker-cluster over-segmentation, residual label noise, and the utilization of cross-view-disagreement samples within a unified iterative framework.

III. Problem Formulation and Framework Overview

A. Problem Formulation

Let $U = \{x_i\}_{i=1}^N$ denote an unlabeled speech corpus, where no speaker identity annotation is available during training. The goal of unsupervised speaker verification is to learn a speaker encoder $f_\theta(\cdot)$ that maps an utterance x_i into a speaker-discriminative embedding

$$z_i = f_\theta(x_i). \tag{1}$$

Since ground-truth speaker labels are unavailable, iterative unsupervised systems usually construct a pseudo-label set

$$\mathcal{Y} = \{y_i\}_{i=1}^N \tag{2}$$

through clustering and use it as supervision for training $f_\theta(\cdot)$.

However, pseudo-labels generated by clustering are not uniformly reliable. We distinguish two error modes. The first is *assignment noise*: the pseudo-label assigned to an utterance may not correspond to its latent speaker identity. The second is *speaker-cluster over-segmentation*: utterances from one speaker may be distributed across multiple pseudo-label classes. Once such pseudo-labels are used as hard supervision, their errors can be reinforced in subsequent refinement iterations.

We use *pseudo-label uncertainty* to describe uncertainty in the assigned pseudo-label, rather than an intrinsic property of the speech sample. Because ground-truth labels are unavailable, this uncertainty cannot be observed directly. Instead, CGPL uses cross-model assignment stability as an operational proxy. After aligning two clustering partitions, a sample is defined as *cross-view consistent* if both ensemble members assign it to the same aligned cluster, and *cross-view discrepant* otherwise. Thus, discrepancy indicates a model-dependent pseudo-label assignment, but does not imply that the sample itself is inherently uncertain or that its pseudo-label is necessarily incorrect.

Based on these definitions, we decompose the learning problem into two reliability-aware subproblems: constructing a pseudo-label subset supported by ensemble consensus and robustly utilizing samples with and without such support.

Specifically, we consider the following two subproblems:

1) **Reliable pseudo-label construction**, which aims to identify a reliable subset

$$U_r \subseteq U, \tag{3}$$

whose pseudo-labels are more likely to preserve speaker consistency.

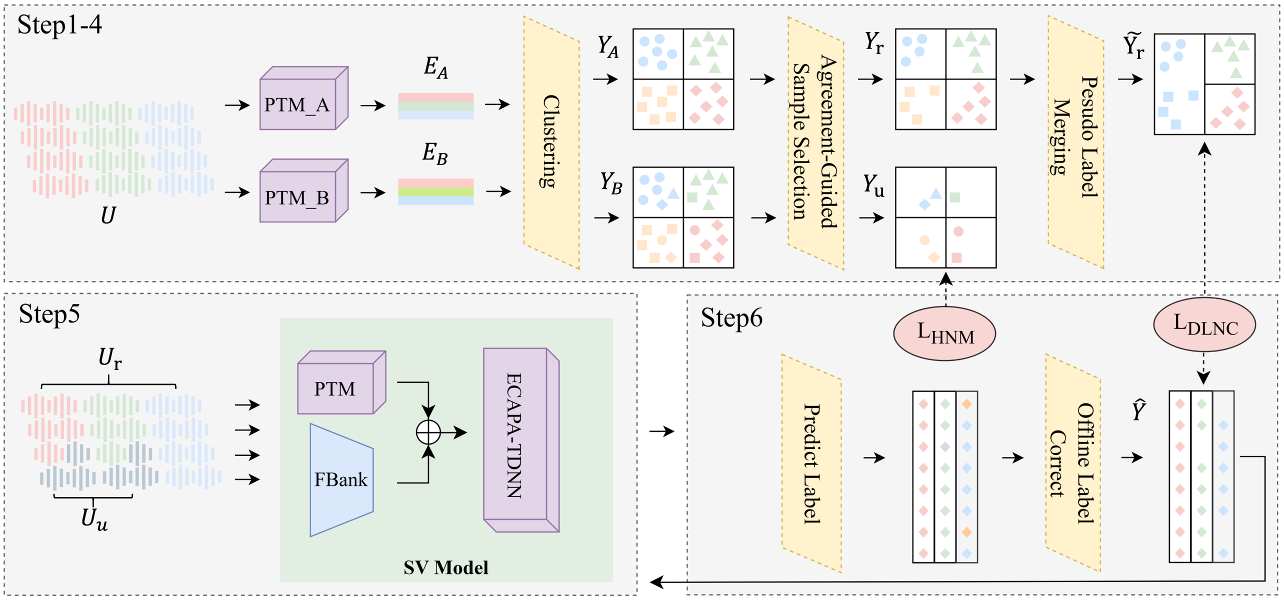

Fig. 1. The overall architecture of the proposed CGPL framework. Steps 1–4 perform consensus-guided reliable pseudo-label construction using two speech pre-trained models, including cluster-label alignment, consensus-based reliable sample selection, and pseudo-label merging. Steps 5–6 conduct noise-robust utilization of consensus-selected and cross-view-disagreement samples through dynamic label-noise-aware supervision and hard-negative-based contrastive learning.

2) **Noise-robust sample utilization**, which aims to optimize the speaker encoder by assigning different learning objectives to the consensus-selected subset U_r and the cross-view-disagreement subset

$$U_d = U \setminus U_r. \tag{4}$$

B. Framework Overview

The proposed framework follows the above decomposition and consists of two sequential stages built on a standard iterative pseudo-label training pipeline.

In the first stage, a heterogeneous ensemble of two speech pre-trained models independently produces two clustering partitions. Since cluster indices are permutation-invariant, the partitions are aligned before their decisions are compared. Samples with identical aligned assignments form the consensus-selected subset. A pseudo-label merging strategy then merges highly similar clusters to alleviate speaker-cluster over-segmentation.

In the second stage, the selected reliable samples are used for pseudo-label-guided supervised training with dynamic label-noise-aware supervision. The original pseudo-label serves as the primary supervision anchor, while the model prediction provides an auxiliary supervision signal. The contribution of this auxiliary signal is adaptively controlled according to both the training-stage reliability and the sample-level posterior margin.

Cross-view-disagreement samples are not discarded. Instead, they are utilized for self-supervised contrastive learning. Positive pairs are generated from different segments of the same utterance, whereas hard negatives are mined from other utterances within the same mini-batch.

Overall, the proposed framework partitions unlabeled data according to cross-model support for their pseudo-labels and assigns dedicated learning objectives to consensus-selected and cross-view-disagreement samples. This design limits hard pseudo-label supervision to cross-model-consistent assignments while retaining the remaining samples for label-free representation learning.

C. Base Iterative Learning Pipeline

For clarity, this subsection describes the standard system components on which CGPL is built; these components are not claimed as methodological contributions.

1) Heterogeneous Ensemble Partition Generation: Speech self-supervised pre-trained models encode speaker information at different Transformer layers. Following prior work [50], an intermediate layer is selected empirically from each model as described in Section VI-A, and its utterance-level representations are used for clustering. We consider wav2vec 2.0, HuBERT, UniSpeech-SAT, and WavLM as candidate ensemble members. Two members are used in the reported system, but the formulation is based on heterogeneous ensemble decisions rather than on a particular PTM pair.

For each ensemble member, utterances are represented as graph nodes and cosine similarity is used to construct weighted edges. Infomap [52], which has been used in speaker clustering and unsupervised speaker verification [36], [53], independently partitions the two graphs. This produces two pseudo-label assignments, Y_A and Y_B, that serve as inputs to the proposed consensus estimator in Section IV-A.

2) Speaker Encoder and Iterative Refinement: Following [49], the downstream speaker encoder combines adaptively fused multi-layer PTM representations with FBank features processed by a convolutional front end. The fused representation is passed to an ECAPA-TDNN backbone [7] for speaker embedding extraction. Both learning objectives introduced in Section V update this shared encoder.

Between training rounds, we adopt the offline pseudo-label refinement procedure of [36]. Each utterance is divided into M segments and the previously trained encoder predicts one label per segment. If the modal prediction Y_t occurs I times and $I/M > 0.5$, the utterance label is updated to Y_t; otherwise, the utterance is omitted from pseudo-label supervision in the next round. This inherited refinement operation is distinct from the proposed objective-level noise-aware supervision.

IV. Consensus-Based Pseudo-Label Construction

Given the two ensemble partitions produced by the base pipeline in Section III-C, this section introduces the proposed pseudo-label construction method. Cross-model consensus is evaluated at the assignment level after partition alignment. Agreement is therefore the concrete decision rule, whereas consensus denotes the ensemble evidence represented by that agreement.

The proposed construction process consists of two steps:

1) **Consensus-based sample selection**, where the two clustering partitions are first aligned using the Hungarian algorithm, and only samples with consistent aligned assignments are retained for pseudo-label supervision;
2) **Pseudo-label merging**, where highly similar pseudo-label clusters are merged to alleviate speaker-cluster over-segmentation while limiting unsupported cluster fusion.

The first step filters model-dependent pseudo-label assignments, and the second reduces over-segmentation among highly similar speaker clusters.

A. Consensus-Guided Sample Selection

Pseudo-label quality in iterative unsupervised speaker verification is sensitive to clustering errors, especially during the initial training stage. Directly using all clustering assignments as hard supervision may introduce inaccurate optimization signals and affect subsequent model refinement. We introduce consensus-guided sample selection (CGS) to estimate whether a pseudo-label assignment is stable across the heterogeneous ensemble. CGS operates on two clustering partitions generated from different speech pre-trained models and does not require speaker annotations or calibrated class probabilities.

The CGS strategy exploits the consistency between two PTM-based clustering assignments. After cluster-label alignment, a sample is selected for pseudo-label-supervised training only when its assignments agree across the two clustering partitions. Samples with inconsistent assignments form the cross-view-disagreement subset and are excluded only from pseudo-label classification; they remain available to the label-free objective in Section V-B.

Specifically, let Y_A and Y_B denote two pseudo-label assignments obtained by clustering the embeddings extracted from two different pre-trained models. The corresponding cluster sets are defined as

$$C_A = \{c_1^A, c_2^A, \dots, c_m^A\}, \tag{5}$$

$$C_B = \{c_1^B, c_2^B, \dots, c_n^B\}, \tag{6}$$

where c_i^A and c_j^B denote the cluster identities in Y_A and Y_B, respectively.

Since clustering labels are permutation-invariant, the numerical indices of clusters cannot be compared directly [54]. Therefore, we first compute the overlap score between clusters from the two partitions:

$$S(i,j) = \left|\left\{x_k \mid x_k \in U, Y_A(x_k) = c_i^A, Y_B(x_k) = c_j^B\right\}\right|, \tag{7}$$

where U denotes the unlabeled training corpus and x_k represents the k-th speech sample.

The Hungarian algorithm [55] is then employed to obtain the maximum-overlap one-to-one cluster alignment:

$$\mathcal{M} = \arg\max_{\pi} \sum_{(i,j)\in\pi} S(i,j), \tag{8}$$

where π denotes a feasible cluster matching configuration. When the two partitions contain different numbers of clusters, the overlap matrix is padded with zero-valued dummy entries. A cluster matched to a dummy entry is treated as unmatched and none of its samples can satisfy the consensus criterion. Based on the obtained matching relationship, the pseudo-labels in Y_B are aligned with those in Y_A:

$$\mathrm{Map}(c_j^B) = c_i^A, \quad \forall (i,j) \in \mathcal{M}, \tag{9}$$

and the aligned pseudo-label assignment is defined as

$$Y_B^{\mathrm{align}}(x_k) = \begin{cases} \mathrm{Map}(Y_B(x_k)), & Y_B(x_k) \text{ is matched,} \\ \varnothing, & \text{otherwise.} \end{cases} \tag{10}$$

After alignment, the agreement indicator for each sample is defined as

$$a_k = \begin{cases} 1, & Y_A(x_k) = Y_B^{\mathrm{align}}(x_k), \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

The consensus-selected subset and the cross-view-disagreement subset are then defined as

$$U_r = \{x_k \in U \mid a_k = 1\}, \quad U_d = U \setminus U_r. \tag{12}$$

The selected pseudo-label assignment for samples in U_r is obtained from the reference partition:

$$Y_r(x_k) = Y_A(x_k), \quad x_k \in U_r. \tag{13}$$

For samples satisfying $a_k = 1$, Y_A and Y_B^{align} provide identical labels after alignment; therefore, Y_A is used as the reference label space.

B. Pseudo-Label Merging

After consensus-guided sample selection, cluster over-segmentation may still exist in the selected pseudo-label set. Specifically, utterances from the same speaker may be divided into multiple pseudo-label clusters due to acoustic variability, recording conditions, and imperfect clustering boundaries. Such fragmentation increases the number of pseudo-label classes and may introduce redundant supervision during subsequent model training.

To alleviate this problem, we apply a centroid-based pseudo-label merging strategy to the selected pseudo-label set. For the i-th pseudo-label cluster, its centroid is computed as

$$\mathbf{c}_i = \frac{1}{N_i}\sum_{k=1}^{N_i}\mathbf{e}_{i,k}, \tag{14}$$

where N_i denotes the number of samples in the i-th cluster and $\mathbf{e}_{i,k}$ represents the embedding of its k-th sample.

The similarity between clusters i and j is measured by the cosine similarity between their centroids:

$$s_{i,j} = \cos(\mathbf{c}_i, \mathbf{c}_j). \tag{15}$$

Instead of using a dataset-independent absolute threshold, we define the merging threshold according to the maximum inter-cluster similarity in the current pseudo-label set:

$$\theta_{\mathrm{PLM}} = \mu \max_{i\neq j} s_{i,j}, \tag{16}$$

where $\mu \in [0,1]$ controls the merging criterion. A larger value of μ results in a stricter merging condition. In all experiments, μ is fixed to 0.97.

Two clusters are merged when

$$s_{i,j} \geq \theta_{\mathrm{PLM}}. \tag{17}$$

The relative threshold adapts the merging criterion to the similarity distribution of the current embedding space. By requiring cluster similarities close to the maximum observed value, PLM reduces cluster fragmentation while limiting unnecessary cluster fusion.

After merging, samples from merged clusters are assigned the same pseudo-label. The resulting pseudo-label set, denoted as $\tilde{Y}_r$, is used as the initial pseudo-label supervision for downstream speaker encoder training.

V. Noise-Robust Utilization by Consensus Status

After consensus-guided pseudo-label construction, the unlabeled corpus is divided into the consensus-selected subset U_r and the cross-view-disagreement subset U_d. The former contains pseudo-labels supported by both ensemble members and is used for pseudo-label-supervised speaker-discriminative training, while the latter is excluded from pseudo-label classification because its aligned assignments differ across the two members.

To utilize these two subsets in a noise-robust manner, we apply dynamic label-noise-aware supervision to U_r and self-supervised contrastive learning to U_d. This design reduces the influence of residual pseudo-label errors while ensuring that cross-view-disagreement samples are not discarded.

Algorithm 1 CGPL: Consensus-Guided Reliable Pseudo-Label Construction

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Input: Unlabeled audio set U, PTM-A, PTM-B, clustering algorithm $\mathcal{C}(\cdot)$, merging coefficient μ
Output: Merged pseudo-labels $\tilde{Y}_r$, consensus-selected subset U_r, cross-view-disagreement subset U_d
Step 1: Heterogeneous ensemble partition generation
Extract embeddings $\mathbf{E}_A$ and $\mathbf{E}_B$ from PTM-A and PTM-B for all samples in U
$Y_A \leftarrow \mathcal{C}(\mathbf{E}_A), \quad Y_B \leftarrow \mathcal{C}(\mathbf{E}_B)$
Step 2: Consensus-based sample selection
Align Y_B to the label space of Y_A using the Hungarian algorithm
$U_r \leftarrow \{u \in U \mid Y_A(u) = Y_B^{\mathrm{align}}(u)\}$
$U_d \leftarrow U \setminus U_r$
$Y_r(u) \leftarrow Y_A(u)$ for $u \in U_r$
Step 3: Pseudo-label merging
Compute cluster centroids using samples in U_r
Compute $\theta_{\mathrm{PLM}} = \mu \max_{i\neq j}\cos(\mathbf{c}_i, \mathbf{c}_j)$
Merge cluster pairs satisfying $\cos(\mathbf{c}_i, \mathbf{c}_j) \geq \theta_{\mathrm{PLM}}$
Obtain merged pseudo-labels $\tilde{Y}_r$
return $\tilde{Y}_r$, U_r, U_d

A. Dynamic Label-Noise-Aware Supervision

Although consensus-guided pseudo-label construction removes samples with inconsistent cross-model assignments, residual label errors may remain in U_r because agreement between ensemble members is evidence of assignment stability, not a guarantee of correctness. Directly treating all selected pseudo-labels as fully correct may introduce biased supervision and amplify incorrect assignments during iterative training. To alleviate this problem, we introduce Dynamic Label-Noise-Aware Supervision (DLNAS), which incorporates model predictions through a confidence-weighted auxiliary self-training term at the objective level.

DLNAS does not modify the selected pseudo-label: it neither replaces nor updates the label during an optimization step. The original pseudo-label remains the primary supervision target, while the model prediction is used only as an auxiliary target with an adaptive weight. This weight is jointly determined by a global confidence schedule and a sample-level posterior-margin confidence score. Consequently, early low-confidence predictions and predictions with poorly separated class posteriors have limited influence on the objective.

For a sample x_i, let

$$P_{i,k} = p(k \mid x_i; \Theta) \tag{18}$$

denote the posterior probability assigned by the current model to class k, where Θ represents the model parameters. The model-predicted label is defined as

$$\hat{y}_i = \arg\max_k P_{i,k}. \tag{19}$$

1) Global Confidence Schedule: The confidence of model predictions varies during training. At the beginning of optimization, the model is still strongly affected by noisy pseudo-label supervision, and its predictions should not dominate the training objective. As training progresses, the learned representation becomes more discriminative, and model predictions can provide more useful auxiliary supervision.

To summarize the model confidence at training step t, we compute the average maximum posterior probability within a mini-batch:

$$\hat{P}_t = \frac{1}{B}\sum_{i=1}^{B} P_{i,\hat{y}_i}, \tag{20}$$

where B denotes the mini-batch size. To reduce batch-level fluctuations, an exponential moving average is adopted:

$$\delta_t = v\delta_{t-1} + (1-v)\hat{P}_t, \tag{21}$$

where $v \in [0,1)$ is the momentum coefficient. The variable δ_t provides a smoothed global confidence schedule that controls the overall strength of prediction-based auxiliary supervision. We use confidence here as an optimization signal and do not interpret it as a calibrated probability that the predicted class is correct.

2) Sample-Level Posterior-Margin Confidence: In addition to the global training state, different samples exhibit different degrees of prediction ambiguity. A sample whose largest posterior probability is only slightly higher than the second-largest probability should contribute less to auxiliary supervision than a sample with a clearly separated prediction.

Let

$$P_{i,\mathrm{2nd}} = \max_{k\neq \hat{y}_i} P_{i,k} \tag{22}$$

denote the second-largest posterior probability for sample x_i. The posterior margin is defined as

$$m_i = P_{i,\hat{y}_i} - P_{i,\mathrm{2nd}}. \tag{23}$$

We normalize the margin by the maximum posterior probability:

$$\omega_i = \frac{m_i}{P_{i,\hat{y}_i}} = 1 - \frac{P_{i,\mathrm{2nd}}}{P_{i,\hat{y}_i}}. \tag{24}$$

Since $P_{i,\hat{y}_i} \geq P_{i,\mathrm{2nd}} \geq 0$, the confidence score satisfies $\omega_i \in [0,1]$. A larger ω_i indicates that the predicted class is more clearly separated from competing classes.

3) Prediction-Margin-Weighted Auxiliary Objective: Given the selected pseudo-label y_i, conventional pseudo-label supervision is formulated as

$$\mathcal{L}_{\mathrm{PL}} = -\frac{1}{B}\sum_{i=1}^{B}\log P_{i,y_i}. \tag{25}$$

This objective assigns the same supervision strength to all selected pseudo-labels and may therefore be affected by residual clustering errors.

To adaptively regulate the contribution of model-predicted auxiliary supervision, we combine the global confidence schedule δ_t and the sample-level posterior-margin score ω_i:

$$\lambda_i = \delta_t \omega_i, \tag{26}$$

where $\lambda_i \in [0,1]$ is the adaptive auxiliary coefficient for sample x_i. The predicted label $\hat{y}_i$ and coefficient λ_i are detached from the computation graph when the auxiliary term is evaluated, preventing the model from reducing the objective by directly manipulating its own weight.

Algorithm 2 Iterative Unsupervised Speaker Verification Training with Consensus-Guided Initialization

Input: Unlabeled audio set U_0, pretrained encoders PTM-A and PTM-B, maximum iterations K
Output: Trained speaker encoder $f_\theta^{(*)}$
Initialize $f_\theta^{(0)}$
Initialization: Consensus-guided pseudo-label construction
$(\tilde{Y}_r^{(1)}, U_r^{(1)}, U_d^{(1)}) \leftarrow \mathrm{CGPL}(U_0)$ {Algorithm 1}
Train $f_\theta^{(1)}$ using

$$\mathcal{L}_{\mathrm{DLNAS}} + \gamma\mathcal{L}_{\mathrm{HNM}}$$

for $k = 2$ to K **do**
  Generate pseudo-labels $\hat{Y}^{(k)}$ from $f_\theta^{(k-1)}$ using the segment-level modal-prediction and majority-acceptance rule in Section III-C
  Train $f_\theta^{(k)}$ on $(U_0, \hat{Y}^{(k)})$ with $\mathcal{L}_{\mathrm{DLNAS}}$
end for
$f_\theta^{(*)} \leftarrow f_\theta^{(K)}$
return $f_\theta^{(*)}$

The proposed DLNAS objective is defined as

$$\mathcal{L}_{\mathrm{DLNAS}} = -\frac{1}{B}\sum_{i=1}^{B}\left[\log P_{i,y_i} + \lambda_i \log P_{i,\hat{y}_i}\right]. \tag{27}$$

The first term preserves the selected pseudo-label as the primary supervision target. The second term introduces model-predicted auxiliary supervision, whose contribution is controlled by both the global training state and the sample-level posterior margin. When the predicted label agrees with the pseudo-label, i.e., $\hat{y}_i = y_i$, the sample-wise loss becomes

$$\mathcal{L}_i = -(1+\lambda_i)\log P_{i,y_i}. \tag{28}$$

In this case, clustering supervision and model prediction support the same target, increasing the effective training weight of the sample.

When $\hat{y}_i \neq y_i$, the selected pseudo-label remains the dominant supervision target, while the predicted label contributes only through the weighted auxiliary term. Since λ_i is small for unstable training stages or low-margin predictions, unreliable predictions are prevented from strongly affecting the optimization.

In Algorithm 2, the consensus-guided initialization and the optimization of $f_\theta^{(1)}$ constitute the first training iteration. Iterations $k \geq 2$ are subsequent refinement iterations based on the previous encoder. The HNM objective is applied to $U_d^{(1)}$ during the first iteration; the subsequent loop uses the accepted pseudo-labels and the DLNAS objective. We use *training iteration* when referring to all three rounds and reserve *refinement iteration* for rounds after the consensus-guided initialization.

B. Contrastive Learning with Hard Negative Mining

The cross-view-disagreement subset U_d is excluded from pseudo-label classification because its cross-model assignments are inconsistent. However, these samples still contain useful speaker-related acoustic information. To exploit these samples without introducing unreliable speaker pseudo-labels, we adopt an utterance-level self-supervised contrastive learning strategy.

For each utterance $x_i \in U_d$, two non-overlapping speech segments are randomly sampled from the same recording and fed into the speaker encoder to obtain embeddings $\mathbf{z}_i^1$ and $\mathbf{z}_i^2$. The two embeddings from the same utterance are treated as a positive pair, while embeddings from different utterances within the same mini-batch are regarded as negative candidates. This formulation provides label-free supervision for cross-view-disagreement samples.

After L_2 normalization,

$$\bar{\mathbf{z}} = \frac{\mathbf{z}}{\|\mathbf{z}\|}, \tag{29}$$

the cosine similarity between two embeddings is calculated as

$$S_{ij} = \bar{\mathbf{z}}_i^1 \cdot \bar{\mathbf{z}}_j^2. \tag{30}$$

The similarity matrix within a mini-batch is represented as

$$S = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1N} \\ S_{21} & S_{22} & \cdots & S_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ S_{N1} & S_{N2} & \cdots & S_{NN} \end{bmatrix}, \tag{31}$$

where N denotes the mini-batch size.

Following a linear transformation,

$$S'_{ij} = \omega S_{ij} + b, \tag{32}$$

the conventional contrastive learning objective is formulated as

$$\mathcal{L}_{\mathrm{CL}} = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{e^{S'_{ii}}}{\sum_{j=1}^{N} e^{S'_{ij}}}. \tag{33}$$

The above objective treats all negative candidates equally. However, in speaker verification, some negative candidates may be acoustically closer to the anchor and therefore provide more challenging discrimination signals. To improve the discriminative capability of the learned speaker representations, we replace the conventional contrastive objective with a hard negative mining (HNM) objective.

For each anchor sample, the hardest negative candidate is selected according to the maximum cosine similarity among all negative pairs:

$$S_i^{\mathrm{hard}} = \max_{j \neq i} S_{ij}. \tag{34}$$

The HNM objective is formulated as

$$\mathcal{L}_{\mathrm{HNM}} = \frac{1}{N} \sum_{i=1}^{N} \max\left(0, S_i^{\mathrm{hard}} - S_{ii} + m\right), \tag{35}$$

where m denotes the margin parameter controlling the separation between positive pairs and hard negative pairs.

Compared with the conventional contrastive objective, HNM explicitly focuses on acoustically confusing negative candidates and encourages larger separation between positive and difficult negative pairs. Since the cross-view-disagreement subset does not provide consensus-supported speaker labels, HNM is used as an auxiliary self-supervised objective rather than a replacement for pseudo-label supervision. Because speaker identities are unavailable during optimization, an utterance selected as a negative can in principle originate from the same speaker as the anchor. The small HNM mini-batch reduces the chance of such collisions but cannot eliminate this false-negative failure mode.

The final optimization objective combines reliable pseudo-label supervision and representation learning on cross-view-disagreement samples:

$$\mathcal{L}_{\mathrm{Total}} = \mathcal{L}_{\mathrm{DLNAS}} + \gamma \mathcal{L}_{\mathrm{HNM}}, \tag{36}$$

where γ controls the contribution of representation learning on cross-view-disagreement samples.

VI. Experimental Setup

A. Datasets

Training Set: To evaluate the proposed CGPL framework, experiments were conducted on the VoxCeleb2 dataset [43], which is a large-scale speaker verification corpus collected from YouTube interview videos. VoxCeleb2 contains 1,092,009 utterances from 5,994 speakers and serves as an extension of the original VoxCeleb dataset. In this work, only the audio modality was utilized, and no speaker identity annotations were used during the entire training process.

Evaluation Set: The proposed method is evaluated on three standard VoxCeleb1 evaluation protocols [60]: Vox1-O, Vox1-E, and Vox1-H. Vox1-O corresponds to the original VoxCeleb1 test set and contains 37,720 verification trials from 40 speakers. Vox1-E extends the evaluation to the entire VoxCeleb1 dataset, consisting of 581,480 verification trials from 1,251 speakers. Vox1-H is a more challenging evaluation protocol containing 552,536 trials from 1,190 speakers, where all speaker pairs share the same nationality and gender, thereby substantially increasing speaker discrimination difficulty.

Performance is evaluated using Equal Error Rate (EER), where the false acceptance rate equals the false rejection rate.

Data Augmentation: To improve robustness under complex acoustic conditions, online data augmentation was employed during training. Specifically, additive noise and reverberation were introduced using the MUSAN [61] and RIR [62] datasets.

B. Implementation Details

For downstream unsupervised speaker verification training, WavLM is selected as the PTM backbone because it provides the strongest speaker discrimination in the layer-wise evaluation. The downstream speaker encoder is based on ECAPA-TDNN with a channel size of 1024.

Training utterances are segmented into non-overlapping 2-second speech chunks. The batch size for pseudo-label-supervised training is set to 512. AdamW is used for optimization together with a triangular cyclic learning-rate schedule. The training procedure contains two stages:

1) The WavLM parameters are first frozen, and only the downstream ECAPA-TDNN is optimized for 10 epochs. The learning rate varies from 1×10^{-8} to 5×10^{-4} under

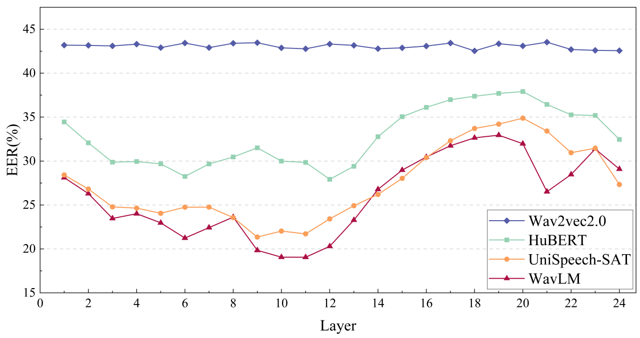

Fig. 2. EER% performance on Vox1-O across different Transformer layers.

a triangular cyclic schedule with a cycle step size of 5 epochs.

2) The WavLM parameters are subsequently unfrozen, and the entire model is jointly fine-tuned for another 10 epochs. The maximum learning rate is reduced to 5×10^{-5} during this stage to stabilize joint optimization.

For contrastive learning with hard negative mining, the mini-batch size is set to 16. The relatively small batch size is intended to reduce, although not completely eliminate, the probability that multiple utterances from the same speaker appear in the same batch and are incorrectly treated as negative pairs. The margin parameter in $\mathcal{L}_{\mathrm{HNM}}$ is fixed to 0.2. During inference, cosine similarity scoring is used without additional score normalization.

C. Evaluation Protocol and Claim Scope

Speaker identities play different roles in optimization and offline analysis. They are not used to construct the CGS agreement indicator, merge pseudo-labels, define the DLNAS or HNM objectives, or update model parameters. They are, however, used in two offline analyses: Vox1-O EER is used to characterize the PTM layers and the contrastive-loss weight, and VoxCeleb2 identities are used to compare clustering configurations and to report pseudo-label quality. Consequently, the reported protocol supports *label-free pseudo-label construction and model optimization*, but it should not be interpreted as fully label-free system development. In particular, Vox1-O is development-informed rather than an untouched test set. Vox1-E and Vox1-H are not used directly for configuration selection, although they belong to the same benchmark family.

All EER and ablation entries are point estimates from the reported runs; repeated-run uncertainty and minDCF are not available in the current experimental record. We therefore interpret small component-wise EER differences as observed changes under this configuration rather than as evidence of statistical significance. Comparisons with published systems are also descriptive because the systems differ in pre-training resources, backbones, clustering procedures, and total computation.

VII. Experimental Results

This section evaluates the proposed framework from both speaker verification performance and pseudo-label construction perspectives. In addition to the main EER comparisons, we analyze pseudo-label quality, the influence of cross-model agreement, and the iterative refinement process. These analyses aim to provide a comprehensive understanding of the proposed consensus-guided pseudo-labeling framework.

TABLE I
Clustering Quality of Initial Pseudo-Label Partitions Generated by Different Clustering Algorithms.

PTM	Method	Precision↑	Recall↑	F1-score↑	NMI↑
WavLM	K-means	0.303	0.209	0.247	0.665
	AHC	0.442	0.346	0.388	0.776
	Leiden	**0.887**	0.237	0.374	0.859
	Infomap	0.795	**0.426**	**0.555**	**0.871**
UniSpeech-SAT	K-means	0.287	0.200	0.236	0.651
	AHC	0.442	0.347	0.389	0.787
	Leiden	**0.901**	0.173	0.291	0.861
	Infomap	0.697	**0.364**	**0.478**	**0.872**

A. Offline Ensemble Member Configuration Analysis

Speaker-related information is unevenly distributed across the Transformer layers of speech pre-trained models. To instantiate the heterogeneous ensemble, we conduct a layer-wise evaluation on Vox1-O to characterize the speaker-discrimination capability of candidate representations. A lower EER indicates that the corresponding layer contains more speaker-discriminative information.

As shown in Fig. 2, WavLM achieves lower EERs than the other evaluated PTMs across most Transformer layers. In particular, the 11th layer of WavLM provides the best verification performance, while the 9th layer of UniSpeech-SAT achieves the second-best result. These two representations are used as the ensemble members for initial pseudo-label generation. As stated in Section VI-C, this use of Vox1-O makes its reported EER development-informed.

Besides their individual verification performance, WavLM and UniSpeech-SAT are trained with different pre-training objectives, which may lead to different representation spaces and clustering behaviors. The reported configuration therefore provides both member competence and representation diversity, two properties commonly associated with effective heterogeneous ensembles.

B. Offline Clustering Configuration Analysis

To select an appropriate clustering algorithm for initial pseudo-label generation, we evaluate K-means, Agglomerative Hierarchical Clustering (AHC) [63], Leiden [64], and Infomap on VoxCeleb2. Speaker embeddings are extracted from the 11th layer of WavLM and the 9th layer of UniSpeech-SAT, which provide the strongest speaker discrimination in the preceding layer-wise analysis. For K-means and AHC, the number of clusters is set to 6,000, which approximately corresponds to the number of speakers in VoxCeleb2. This setting is only used for algorithm comparison, since these two methods require a predefined number of clusters. In contrast, Infomap and Leiden determine the cluster structures without using the ground-truth speaker count. The ground-truth speaker identities are used for this offline configuration comparison but are not input to pseudo-label generation or model optimization. Thus, the selection of Infomap is label-informed even though the subsequent clustering operation itself does not use speaker identities.

Clustering performance is evaluated using Precision, Recall, F1-score, and Normalized Mutual Information (NMI), with $\mathrm{F1} = 2\,\mathrm{Precision}\,\mathrm{Recall}/(\mathrm{Precision} + \mathrm{Recall})$. As shown in Table I, Leiden achieves the highest precision for both PTM feature sets, indicating that it produces highly compact clusters. However, its relatively low recall suggests that many speaker samples are distributed into fragmented clusters. Infomap provides a better balance between precision and recall, leading to the highest F1-score and NMI for both WavLM and UniSpeech-SAT representations. Therefore, Infomap is selected to generate the initial clustering partitions for the subsequent consensus-guided pseudo-label construction.

C. First-Round Pseudo-Label Analysis

To characterize the initial pseudo-label partitions, we additionally report Clustering Accuracy (A_c) and macro-averaged cluster Purity. Clustering Accuracy measures the maximum sample-level agreement between pseudo-label assignments and reference labels after optimal label alignment. Purity measures internal label consistency while assigning equal weight to each pseudo-label cluster, irrespective of cluster size. These metrics use reference identities for offline analysis; the identities are not input to pseudo-label generation or model optimization.

The Clustering Accuracy A_c is defined as

$$A_c = \frac{\max\limits_{\pi} \sum_{(i,j)\in\pi} C(i,j)}{M}, \tag{37}$$

where $C(i,j)$ denotes the number of samples shared by the i-th pseudo-label cluster and the j-th reference label class, π denotes a feasible one-to-one mapping between pseudo-label clusters and reference label classes, and M denotes the total number of evaluated samples. The optimal mapping is obtained using the Hungarian algorithm [55].

Purity is defined as

$$\mathrm{Purity} = \frac{1}{N_c} \sum_{i=1}^{N_c} \frac{\max_j C(i,j)}{\sum_j C(i,j)}, \tag{38}$$

where N_c denotes the number of pseudo-label clusters. For each pseudo-label cluster, the proportion of samples associated with its most frequent reference label is first computed, and the resulting cluster-wise purities are then averaged over all clusters. A higher Purity indicates greater average label consistency within individual pseudo-label clusters. Because this definition is macro-averaged, it is sensitive to the number and size distribution of clusters and should be interpreted together with N_c and N_s.

Using Infomap clustering, two initial pseudo-label partitions, denoted as Y_A and Y_B, are generated from WavLM and UniSpeech-SAT embeddings, respectively. After cluster-label alignment, samples with consistent assignments across the two PTM-based partitions form the consensus-selected subset, whose corresponding pseudo-label set is denoted as Y_r.

TABLE II
COMPARISON OF THE PROPOSED CGPL FRAMEWORK WITH REPRESENTATIVE SINGLE-STAGE UNSUPERVISED SPEAKER VERIFICATION METHODS IN TERMS OF EER (%) ON THE VOX1-O, VOX1-E, AND VOX1-H TEST SETS.

Category	Method	Model	Vox1-O	Vox1-E	Vox1-H
Contrastive	SimCLR + MSE[10]	ResNet34	8.28		
	MoCo + WavAug[11]	TDNN	8.23	-	-
	SimCLR + Margin[12]	ResNet34	7.85	-	-
	SimCLR + AAT[18]	ECAPA-TDNN	7.36	-	-
	C3-MoCo[13]	ECAPA-TDNN	6.40	-	-
	MCL-DPP[19]	ECAPA-TDNN	2.89	3.17	6.27
	SimCLR-SSPS[14]	ECAPA-TDNN	2.57	3.11	5.56
Self-distillation	DINO + Cosine loss[22]	ResNet34	6.16	-	-
	DINO + Curriculum[23]	ECAPA-TDNN	4.47	-	-
	CA-DINO[21]	ECAPA-TDNN	3.59	3.85	6.92
	RDINO[24]	ECAPA-TDNN	3.29	-	-
	EBCA-DINO[25]	ECAPA-TDNN	2.94	2.99	5.14
	DINO-SSPS[14]	ECAPA-TDNN	2.53	2.55	4.93
	DINO + Aug[26]	ECAPA-TDNN	2.51	2.47	4.79
	C3-DINO[13]	ECAPA-TDNN	2.50	-	-
PTM	Sub-PTM[36]	WavLM&ECAPA-TDNN	2.55	-	-
	CGPL(ours)	WavLM&ECAPA-TDNN	**2.33**	**2.37**	**4.28**

As summarized in Table III, the agreement-based selection retains 755,517 out of 1,092,009 training utterances, corresponding to a sample retention rate of 69.19%. On this selected subset, Y_r has higher clustering metrics than either full input partition. Specifically, the F1-score increases from 0.555 and 0.478 to 0.692, the NMI from 0.871 and 0.872 to 0.942, and A_c from 0.569 and 0.519 to 0.677 relative to Y_A and Y_B, respectively. Purity increases from 0.784 and 0.708 to 0.939. These observations show that cross-model agreement identifies a subset with greater reference-label consistency. Because the compared partitions have different sample coverage, the analysis does not isolate agreement from the general effect of selecting an easier subset and is not a coverage-matched comparison against alternative selection criteria.

Pseudo-label merging is subsequently applied to Y_r, resulting in the merged pseudo-label set $\tilde{Y}_r$. Compared with Y_r, $\tilde{Y}_r$ further improves the F1-score from 0.692 to 0.702, the NMI from 0.942 to 0.944, A_c from 0.677 to 0.683, and Purity from 0.939 to 0.941. Meanwhile, the number of pseudo-label classes (N_c) decreases from 18,180 to 17,819, while the number of retained samples (N_s) remains unchanged. The reduction in pseudo-label classes together with the improvement in clustering metrics suggests that pseudo-label merging can alleviate cluster fragmentation without degrading cluster-level consistency.

Based on the above analysis, $\tilde{Y}_r$ is adopted as the initial pseudo-label supervision for downstream speaker verification training. Samples without cross-model consensus form the cross-view-disagreement subset U_d and are further utilized through the self-supervised contrastive learning objective with hard negative mining.

D. Verification Performance

Table II compares the proposed CGPL framework with representative single-stage unsupervised speaker verification methods. After the first training iteration, CGPL achieves EERs of 2.33%, 2.37%, and 4.28% on Vox1-O, Vox1-E, and Vox1-H, respectively, which are the lowest values in Table II. Because the published systems differ in backbone, pre-training resources, and optimization protocol, the table provides a descriptive comparison rather than a controlled ranking. Within the reported configuration, the results are consistent with effective initial optimization using the proposed pseudo-label construction and sample-specific objectives.

TABLE III
QUALITY COMPARISON OF INITIAL PSEUDO-LABEL PARTITIONS.

Label set	F1-score↑	NMI↑	A_c ↑	Purity ↑	N_c	N_s
Y_A	0.555	0.871	0.569	0.784	20385	1092009
Y_B	0.478	0.872	0.519	0.708	21296	1092009
Y_r	0.692	0.942	0.677	0.939	18180	755517
$\tilde{Y}_r$	**0.702**	**0.944**	**0.683**	**0.941**	17819	755517

The performance improvement is consistently observed across Vox1-O, Vox1-E, and Vox1-H. Compared with Vox1-O, Vox1-E and Vox1-H introduce more challenging verification conditions with a larger number of trials and more confusable speaker pairs. The low EERs across these protocols indicate that the learned representations remain effective as the VoxCeleb1 trial lists become more challenging. They do not, however, establish robustness to a new corpus, language, or acoustic domain.

The strong first-round performance is consistent with the pseudo-label quality analysis in Table III. By selecting samples with consistent cross-model assignments and further applying pseudo-label merging, the proposed framework provides a cleaner initial supervision set for speaker encoder optimization. The remaining pseudo-label noise is addressed by the proposed DLNAS objective, while cross-view-disagreement samples continue to contribute through the self-supervised contrastive learning objective with hard negative mining.

Table IV compares the proposed CGPL framework with representative iterative unsupervised speaker verification methods. After three training iterations, CGPL achieves EERs of 1.01%, 1.36%, and 2.53% on Vox1-O, Vox1-E, and Vox1-H, respectively, which are the lowest values among the systems listed in the table.

Compared with the iteration counts reported for the other multi-stage methods, CGPL reaches its best reported values af-

TABLE IV
COMPARISON OF THE PROPOSED CGPL FRAMEWORK WITH REPRESENTATIVE MULTI-STAGE UNSUPERVISED SPEAKER VERIFICATION METHODS IN TERMS OF EER (%) ON THE VOX1-O, VOX1-E, AND VOX1-H TEST SETS.

Method	Stage I	Model	Clustering	Iter	Vox1-O	Vox1-E	Vox1-H
Fully-supervised	-	WavLM&ECAPA-TDNN	-	1	0.94	1.19	2.25
Duke-1[15]	SimCLR	ResNet34	K-Means	5	3.45	4.02	5.57
IDlab[16]	MoCo	ECAPA-TDNN	AHC	7	2.10	-	-
JHU[27]	DINO	Res2Net50	AHC	4	1.89	-	-
Duke-2[17]	SimCLR	ResNet34	K-Means	4	1.81	-	-
LGL[18]	SimCLR	ECAPA-TDNN	K-Means	5	1.66	2.18	3.76
CA-DINO[21]	DINO	ECAPA-TDNN	K-Means	3	1.58	1.88	3.29
DLG-LC[28]	DINO	ECAPA-TDNN	K-Means	5	1.47	1.78	3.19
MCL-DPP[19]	SimCLR	ECAPA-TDNN	K-Means	4	1.44	1.77	3.27
AT + HT[29]	DINO	ECAPA-TDNN	K-Means	5	1.35	-	-
BDS-BPLC[30]	DINO	ECAPA-TDNN	K-Means	5	1.33	1.56	2.78
Co-Meta[20]	SimCLR	ECAPA-TDNN	K-Means	7	1.27	1.82	3.31
Sub-PTM[36]	WavLM	WavLM&ECAPA-TDNN	Infomap	5	1.25	-	-
IPL[65]	i-vector	MFA-Conformer	AHC	8	1.06	2.16	3.61
CGPL(ours)	WavLM&UniSpeech-SAT	WavLM&ECAPA-TDNN	Infomap	3	**1.01**	**1.36**	**2.53**

TABLE V
DETAILED RESULTS FOR EACH TRAINING ITERATION. METRICS ARE REPORTED IN B/A FORMAT, WHERE B DENOTES THE PSEUDO-LABEL SET BEFORE AND A DENOTES THE SET AFTER THE OFFLINE SEGMENT-LEVEL MODAL-PREDICTION REFINEMENT DESCRIBED IN SECTION III-C.

metric	Iter1	Iter2	Iter3
F1-score↑	0.702 / 0.719	0.826 / 0.865	0.883 / 0.907
NMI↑	0.944 / 0.945	0.962 / 0.973	0.976 / 0.981
A_c↑	0.683 / 0.707	0.868 / 0.879	0.916 / 0.924
Purity↑	0.941 / 0.966	0.954 / 0.969	0.974 / 0.976
N_c	17819 / 14628	11501 / 9447	11119 / 8820
N_s	755517 / 991261	1092009 / 1049500	1092009 / 1068949

ter three training iterations. The observed progression suggests that a higher-consistency initial pseudo-label construction can facilitate subsequent representation refinement and reduce the influence of noisy supervision during iterative training.

Unlike conventional iterative approaches that mainly rely on repeated pseudo-label updates, the proposed framework introduces agreement-based selection before optimization and assigns different learning objectives to consensus-selected and cross-view-disagreement samples. The obtained results indicate that improving pseudo-label reliability at the initialization stage is beneficial for stable iterative unsupervised speaker verification.

E. Iterative Pseudo-Label Refinement

Table V summarizes the evolution of pseudo-label quality and training data coverage during iterative optimization. Across the three training iterations, F1-score, NMI, A_c, and Purity consistently improve, indicating that the pseudo-label assignments become progressively more consistent with the underlying speaker structure while also exhibiting higher internal label consistency. These improvements are observed both before and after offline pseudo-label refinement and document the evolution of the pseudo-label sets throughout iterative training.

During the first iteration, consensus-based selection retains only samples with consistent cross-model assignments, resulting in a smaller but higher-quality supervision subset. As the speaker encoder is optimized with this cleaner supervision, the learned representations become more discriminative, allowing additional utterances to be incorporated during subsequent pseudo-label refinement. Accordingly, N_s increases from 755,517 in the initial reliable subset to over one million samples in later iterations. This trend indicates that the proposed initialization strategy does not permanently discard cross-view-disagreement samples, but progressively enables their utilization as the quality of the learned representation and pseudo-label structure improves.

At the same time, the number of pseudo-label classes generally decreases during refinement, while Purity continues to increase. This joint trend suggests that the iterative process not only expands sample coverage, but also gradually alleviates cluster fragmentation without sacrificing the internal consistency of individual pseudo-label clusters. Although the final number of pseudo-label classes remains larger than the actual number of speakers, the verification performance indicates that moderate over-segmentation does not prevent effective speaker representation learning.

The improvement becomes smaller in later iterations, indicating that the pseudo-label assignments and speaker representations gradually approach a more stable state. Overall, the iterative results are consistent with consensus-guided initialization providing a useful starting point for subsequent pseudo-label refinement in the evaluated VoxCeleb setting.

F. Ablation Study and Contrastive Loss Analysis

Table VI presents the ablation results on Vox1-O after the first training iteration. All variants are evaluated under the same training condition to analyze the contribution of each component before iterative pseudo-label refinement. The progressive reduction in the reported EER is consistent with complementary contributions from pseudo-label construction and the two training objectives. As noted in Section VI-C, these point estimates do not establish statistical significance for the smaller component-wise differences.

Starting from the baseline pseudo-label supervised training, consensus-based sample selection provides a substantial performance improvement. This indicates that improving the quality of the initial supervision is particularly important

TABLE VI
COMPONENT-WISE ABLATION RESULTS ON VOX1-O IN TERMS OF EER (%).

$\mathcal{L}_{PL}$	CGS	PLM	$\mathcal{L}_{DLNAS}$	$\mathcal{L}_{CL}$	$\mathcal{L}_{HNM}$	EER (%)
✓						2.776
✓	✓					2.431
✓	✓	✓				2.415
	✓	✓	✓			2.384
	✓	✓	✓	✓		2.369
	✓	✓	✓		✓	2.334

during the early optimization stage. Pseudo-label merging further reduces the EER, suggesting that reducing cluster fragmentation provides additional benefits after consensus-based selection.

Replacing the conventional pseudo-label objective with DLNAS reduces the reported EER from 2.415% to 2.384%. This observation is consistent with the prediction-guided auxiliary term changing the contribution of samples under residual pseudo-label noise while retaining the original pseudo-label as the primary target. The ablation does not by itself determine whether the gain arises from noise mitigation, confidence-based reweighting, or another optimization effect.

The two contrastive learning variants further reduce the reported EER, showing that the samples excluded from pseudo-label supervision can contribute useful training signals under the evaluated configuration. Compared with the standard contrastive objective, HNM achieves a lower EER, suggesting that emphasizing acoustically challenging negative pairs provides additional discriminative information for speaker representation learning.

Overall, the ablation results indicate that the proposed components address different aspects of unsupervised speaker verification. Consensus-based selection and pseudo-label merging improve the quality of pseudo-label supervision, while DLNAS and HNM enhance robustness during model optimization. Their observed effects are complementary in the sequential ablation, but the small differences for PLM, DLNAS, and conventional contrastive learning should be interpreted cautiously in the absence of repeated-run uncertainty.

Fig. 3 further investigates the influence of the contrastive loss weight γ on Vox1-O performance after the first training iteration. For both $\mathcal{L}_{\mathrm{CL}}$ and $\mathcal{L}_{\mathrm{HNM}}$, the best performance is obtained at $\gamma = 0.2$, while smaller or larger weights lead to degraded EERs. This result indicates that an appropriate balance between pseudo-label supervision and contrastive learning on cross-view-disagreement samples is important during early-stage optimization.

When $\gamma = 0.1$, the contribution of contrastive learning on cross-view-disagreement samples is limited, resulting in relatively smaller performance gains. Increasing γ to 0.2 provides the largest improvement, showing that moderate contrastive supervision effectively complements the pseudo-label-based objective. However, excessively large weights ($\gamma = 0.5$ or 1.0) degrade performance for both contrastive objectives, possibly because the contrastive objective begins to dominate the optimization and reduces the relative contribution of reliable pseudo-label supervision.

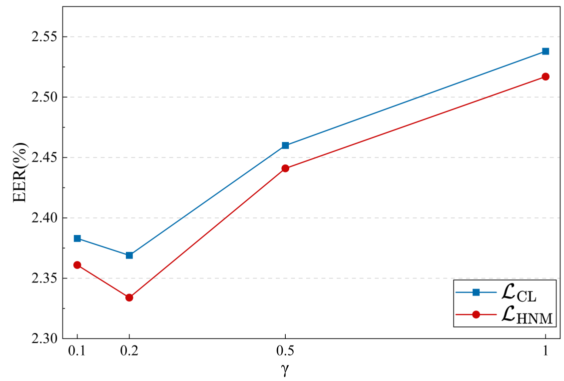

Fig. 3. Effect of contrastive loss weight γ on Vox1-O EER after the first training iteration.

These results show that both the contrastive objective design and its weighting strategy are associated with the utilization of cross-view-disagreement samples under the reported Vox1-O-informed configuration.

VIII. DISCUSSION AND LIMITATIONS

The central observation of this study is that agreement between two aligned PTM-based clustering partitions identifies an initial subset with greater reference-label consistency than either full input partition. This result supports cross-model assignment stability as a practical signal when ground-truth labels cannot be used during optimization. The verification results and sequential ablation further show that the complete pipeline can use this subset to initialize iterative training while retaining the cross-view-disagreement samples through a label-free auxiliary objective.

The pseudo-label analysis does not, however, establish that agreement is uniquely responsible for the higher clustering metrics. CGS retains 69.19% of the training utterances, whereas the two input partitions cover the full corpus. The observed difference may therefore reflect both the consensus criterion and the more general effect of selecting an easier subset. In the same way, the sequential ablation associates PLM, DLNAS, and HNM with lower point-estimate EERs but does not isolate every mechanism. In particular, the DLNAS result is consistent with both residual-noise mitigation and confidence-dependent sample reweighting.

The experimental protocol imposes additional boundaries. Vox1-O informed the PTM-layer and contrastive-weight analyses, and VoxCeleb2 identities informed the offline clustering comparison. The reported study therefore evaluates label-free pseudo-label construction and gradient optimization, not a fully label-free model-selection pipeline. Moreover, the comparison systems use different backbones, pre-training resources, and iteration budgets, and the present results do not include repeated-run uncertainty, minDCF, or total computational cost. These factors limit claims about statistical significance, controlled state-of-the-art performance, and efficiency.

CGPL also inherits several methodological limitations. One-to-one Hungarian alignment may reject potentially compatible clusters when the two partitions have different granularities. The relative PLM threshold depends on the maximum observed inter-cluster similarity and may therefore be sensitive to an atypical cluster pair. Finally, HNM cannot identify same-speaker utterances without labels and may select them as false negatives. Because all reported training and evaluation data belong to the VoxCeleb benchmark family, the findings should not be extrapolated to cross-lingual, cross-corpus, or strongly shifted acoustic conditions without additional evaluation.

IX. Conclusion

This paper presented CGPL, a heterogeneous-ensemble framework that uses aligned clustering agreement to select initial pseudo-labels and assigns different objectives to consensus-selected and cross-view-disagreement samples. On VoxCeleb2, the selected subset retained 69.19% of the utterances and had higher pseudo-label consistency than either input partition. After three training iterations, the resulting system achieved EERs of 1.01%, 1.36%, and 2.53% on Vox1-O, Vox1-E, and Vox1-H, respectively.

Within the reported protocol, these findings support cross-model agreement as an operational signal for initializing iterative unsupervised speaker verification. This conclusion is limited to label-free pseudo-label construction and model optimization in the VoxCeleb setting; configuration selection used offline label-informed analyses, and cross-domain behavior was not evaluated. Future work should test consensus-guided selection under independent development protocols and distribution shifts, and should examine alignment granularity, false negatives, and computational cost.

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